3RD HARSH-ENVIRONMENT MASS SPECTROMETRY WORKSHOP

Regolith Evolved Gas Analyzer (REGA): An Instrument to Characterize the Martial Soil Mineralogy and Atmospheric Composition

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ABSTRACT

The Regolith Evolved Gas Analyzer (REGA), a high temperature furnace, soil handling/metering system and mass spectrometer instrument, for measuring and determining the reactivity and mineralogical composition of soil samples was develop for a future Mars lander mission. REGA would provide key mineralogical and reactivity data that is needed to understand the soil chemistry, to identify locations of possible ancient life on Mars, and to determine potential hazards to human explorers and equipment. REGA is capable of conducting measurements directly on soil samples which include: mass spectrum analysis of evolved gases from heated martian soil samples from Mars ambient temperature to 900?C, and identification of liberated chemicals, e.g. oxygen, sulfur, chlorine, and fluorine. REGA is also capable of long term chemical and isotopic composition monitoring of the martian atmosphere. The REGA instrument consists of four primary components: a flight proven mass spectrometer, a high temperature furnace, a microprocessor, and a soil handling system. Soil samples are gathered by an external arm containing a scoop or drill. A sample is place in the inlet orifice where the fines are sifted into a metering volume and subsequently dropped into a crucible placed under the metering volume. A movable arm then places the crucible in the furnace. The furnace is closed sealing the inner volume containing the crucible from the martian atmosphere. A heating cycle is initiated. As the soil sample is heated through a programmed pattern, the gases evolved at each temperature are passed through a transfer tube to the mass spectrometer for analysis and identification. The REGA Mass Spectrometer (MS) is a double focussing magnetic-sector mass spectrometer. Major components of the instrument are a transfer tube from the furnace or the atmospheric gas inlet, an inlet valve and microleak, an ion source, an electrostatic analyzer, a magnetic analyzer and two ceramic electron multipliers, one each for the high and low mass ranges. Gases to be analyzed pass into the ion source though the microleak. The ionizing electron beam is produced by thermal emission from one of two small tungsten filament; the second filament is for redundancy. The filament and associated electronics each require 1 watt of power. Alternative cold electron sources are being investigated. The ion source is controlled by a microprocessor. Two emission currents (20µA and 200µA) and four electron energies (from 20eV to 75eV) can be selected. This capability extends the mass spectrometer's dynamic range by a factor of 10. It also provides extra information on the molecular constituents and atomic species, by varying their cracking patterns and charge states. For example, CO2 spectrum has very high mass peaks at (besides the parent peak at 44 Da) 28 Da (CO), 22 Da (doubly charged CO2), 16 Da(O) and 12 Da (O). Using 22 eV electrons, all dissociated peaks are below 1 % of the parent. This extended dynamic range and additional information on molecular constituents will facilitate and greatly simplify analysis of the mass spectrum data. The REGA system parameters along with data from sample test runs will be presented.